Furthermore, these self-aggregators may continue to enjoy a latency advantage over customers of competing consolidators.

To the extent that the changes to proprietary feed subscriptions described above are realized, the exchanges would have corresponding losses in revenue or profit from the provision of proprietary data. Since the Commission is unable to determine how many broker-dealers or other market participants would no longer want to use proprietary data feeds as a result of this rule, it is unable to determine the size of this potential reduction in revenue or profit.

One commenter stated that if the exchanges' revenues from market data are reduced, the price of trading services would likely increase, because the loss of revenue "will have to be offset."[2392] The Commission disagrees with this commenter because a reduction in total revenue in and of itself does not necessarily make it optimal for a firm to increase its prices. The Commission expects that prices are set to optimize the amount of profit the firm can extract from the market, and given that this has been done an increase in prices today would not increase profit. A reduction in revenue by itself does not change any of these considerations. All firms must balance a loss in customers against an increase in the revenue received per customer when considering a price increase, and in order for it to be optimal to increase prices, something about this balance must change. Thus, the Commission does not believe that a reduction in total revenue for exchanges will necessarily make it optimal for them to adjust any of their fees, including fees for trading services.

This commenter also added that this scenario of increases in trading fees would follow "if the all-in price of trading is already at the competitive level…."[2393] It is not clear that this

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[2392] See Nasdaq Letter IV at 30.

[2393] See Nasdaq Letter IV at 30.

assumption is met in the market today. The Commission has discussed above the competition that exists in the market for trading services,[2394] and separately, discussed indicia that the market for proprietary data may not be subject to robust competition.[2395]

To the extent that exchanges would find it profitable to increase their trading fees following the implementation of this rule, the Commission believes that the market for trading services is subject to competition, as discussed above, and, as a result, any potential for fees to increase will be constrained by this competition.[2396]

A commenter stated that without profit from selling market data, exchanges would lack the funds necessary to finance improvements to their trading systems, including innovations in order types.[2397] The Commission disagrees because it believes the exchanges only fund improvements and innovations in their trading businesses that have a positive net present value, because this would be consistent with the behavior of any firm seeking to maximize profit. While the Commission acknowledges that alternative sources of funding to internally held cash may be more expensive (or less convenient) sources of financing, the Commission nevertheless believes that the exchanges will continue to be able to finance their best investment opportunities, which are the same projects the exchanges finance today. This is because such opportunities will represent a profit opportunity to both the exchanges and potential sources of financing.

2394 See supra Section V.B.3(d) for a discussion of competition in the market for trading services.

2395 See supra Section V.B.3(b) for a discussion of the market for proprietary data products.

2396 See supra Section V.B.3(d).

2397 See Nasdaq Letter IV at 50.

(b) New Entrants into the Market Making, Broker-Dealer and other Latency Sensitive Trading Businesses

The Commission believes that the final amendments may lead to new market participants entering the market making, smart order routing broker-dealer, and other latency sensitive trading businesses.[2398] This is because the final amendments may help to reduce the information asymmetries between those who choose to rely on proprietary data feeds and those who rely on the feeds from the exclusive SIPs.[2399] For instance, it is possible that currently there are broker-dealers who might want to compete in the business of sophisticated order routing but choose not to because of the cost of the market data necessary to be competitive. To the extent that the final amendments make consolidated market data a viable data product for smart order routing, the Commission believes that these changes could induce these broker-dealers to enter the business.[2400] This would have the benefit of increasing competition in the sophisticated order routing broker-dealer business.

The Commission believes that access to this new, faster consolidated market data could encourage new entrants into the automated market maker business. This would not only improve the competitiveness of this business but also may increase liquidity in the corresponding markets.

2398 One commenter stated that the rule would encourage participation in equity markets. See IEX Letter at 9.

2399 One commenter said it would enhance competition, although not completely eliminate the two-tiered structure of the data market. See Virtu Letter at 2.

2400 These would be broker dealers who have not entered these businesses because, currently, the only way to obtain the benefits associated with the new, expanded core data and decentralized consolidation model is to subscribe to proprietary data feeds.

If these new entrants use a competing consolidator, and if a small latency differential between competing consolidator feeds and the proprietary data feeds remains, then this effect of encouraging new entrants is likely to be small.[2401] If instead these potential new entrants were to become self-aggregators, then this effect of encouraging new entrants is not likely to be small, because the Commission believes that there is unlikely to be a significant latency differential between being a self-aggregator and using proprietary data feeds.[2402] However, if self-aggregation is required to be a new entrant in these businesses, the number of potential new entrants could be small, since using a competing consolidator may be more convenient or less costly than self-aggregating.[2403] It is also possible that potential participants in the sophisticated SOR, automated market making, and other latency sensitive trading businesses may find that they cannot compete effectively without using the data that would remain exclusive to proprietary feeds. To the extent this is the case, the effects discussed above would be further limited.

One commenter stated that the final amendments would create new information asymmetries because of the possibility that competing consolidators could customize their products, which would lead to differences in information between their customers. This commenter stated that this is in contradiction to the claim that information asymmetries will be reduced.[2404] The Commission does not believe that the possibility of a reduction in information

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2401 See supra Section V.B.2(b).

2402 This is because the Commission believes that self-aggregators will use substantially the same technology and methods to perform the self-aggregation function, including the same vendors for such technology, as are used today by those market participants who aggregate the proprietary data feeds.

2403 For related discussion on latency advantages, see supra note 2391.

2404 See Nasdaq Letter IV at 48.

asymmetry in the market is negated by the potential for product differentiation by competing consolidators. New consolidated market data, aggregated in a decentralized consolidation model, will present the opportunity for improvement in the quality of market data received today for those market participants capable of exploiting these improvements. These improvements are relative to the exclusive SIP feeds today. For these market participants who can take full advantage of expanded core data and the decentralized consolidation model, the improvements to their utilization of market data are likely to be more significant than the differences that might emerge between competing consolidators product offerings that improve over the current exclusive SIP feeds. Thus, such market participants will represent a reduction in information asymmetries between users of core data and users of proprietary data. On the other hand, the Commission believes that those market participants who elect to use any low cost, or display feed, options offered by competing consolidators are likely to be participants who currently do not make use of sophisticated market data access. Thus, for these market participants, information asymmetries with respect to latency will be no worse than they are currently, though these market participants may still benefit from the expanded content.[2405]

In addition, many of the differences between competing consolidator products (and their use by market participants) will be driven by differences in what those market participants find most useful for their trading needs, and differences in the ability to process and take advantage of new consolidated market data products distributed by competing consolidators, and these differences exist today.[2406]

2405 See supra Section V.C.1 for a discussion of the benefits of the expanded content of core data to market participants.

2406 See supra Section V.B.2(c).

One commenter stated that the decentralized consolidation model would exacerbate the differences in advantage and information between market participants and perpetuate a "multi-tiered" market structure. The commenter pointed out the likelihood that different competing consolidators would likely develop products with different levels of performance and charge different prices for them. This commenter concluded that this would result in the promotion of even more tiers of separation in market data access than the two tiers separating those who can afford proprietary data and those who cannot.[2407] Commenters also stated the rule would not reduce the difference that currently exists between those who access market data in a fast, sophisticated manner and those who do not. Specifically, these commenters stated that the self-aggregator option available in the final rules will enable the advantages of the fastest users of market data to remain, because these self-aggregators would inevitably have a significant speed advantage over competing consolidators.[2408]

The Commission disagrees with these commenters, and believes that the rule will reduce the differences between existing tiers of market data access, and that the self-aggregator option is essential in producing this outcome. This is because the distinctions between market data access capabilities that exist today are driven by more than just the price and availability of data (as

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2407 See Nasdaq Letter IV at 8 ("Finally, the Commission ignores the likelihood that different consolidators will provide differing levels of service, replacing an allegedly two-tiered market with a multi-tiered market. Even if multiple competing consolidators end up racing against each other to produce unique or superior data products or to distribute data more quickly, they would likely charge premiums for better products and faster services. If so, whatever concerns the SEC may have now about market participants needing to pay high costs to access the best and fastest data will not be solved by its Proposal; to the contrary, the Proposal would only make this problem worse.").

2408 See, e.g., FINRA Letter at 8, NYSE Letter II at 23.

explained further below), and so to the extent such differences remain they will not be a result of these rules. Additionally, the final rules will likely reduce one of the key cost barriers for market participants interested in self-aggregation, thereby reducing the advantage held by those market participants that can afford and choose to pay for it today.

In the context of market data access broadly, there exist many differences in the approaches taken to obtain, process, and use market data.[2409] These differences arise because of differentiation across many aspects of the market data access processes, and the Commission does not expect these differences to go away as a result of these rules. Furthermore (and as explained above[2410]), it is the Commission's understanding that some of these differences exist because the strategies employed by market participants do not all require exactly the same level of sophistication in market data processing to run effectively, as such some participants will be unwilling to change how they consume real time data even if given the opportunity to do so.[2411] What this means is that any discussion of multiple tiers of market data access must be understood within the context of the complex differences in data use across market participants that exist today.

At the same time, there are market participants, within each of the levels of sophistication for market data access described above,[2412] who may be able to significantly improve their

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2409 See supra Section V.B.2(c), where additional details of the various approaches are described.

2410 See supra Section V.B.2(c).

2411 For example, the Commission believes that retail investors have no need for sub-millisecond improvements in latency, but do need timely and complete market data in order to make investment decisions.

2412 See supra Section V.B.2(c).

ability to compete as a result of the rule, both at their current level of capability and beyond. This is because core data will now be delivered according to the decentralized consolidation model, which introduces an incentive structure that will likely result in improvements to latency; and because core data will now contain additional content.[2413] To the extent that these rules result in greater affordability of high quality market data, firms may find they are able to use the savings obtained from substituting away from proprietary feeds to invest in the technology and personnel necessary to increase the level of sophistication at which they use market data. These investments may take the form of purchasing the highest quality, lowest latency aggregation technology from a competing consolidator (which may be priced at a premium compared to lower performing products) or investing in the infrastructure necessary to self-aggregate.[2414] In either case, market participants have a greater opportunity to improve their quality of market data access, and therefore, the competitiveness at each level of market participation may be increased.

Also, the Commission does not believe that the self-aggregator option will further solidify the advantages held by sophisticated users of market data. To the contrary, the Commission continues to believe that the self-aggregator option assists in promoting the ability of a wider array of market participants to improve their access to market data. The Commission believes that the advantages of self-aggregators today come in part from the significant costs to

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2413 Furthermore, the Commission believes it is likely that at least some competing consolidators will provide all core data in their product offerings. See supra Section V.C.1(c) for additional discussion of this point.

2414 It is the Commission's understanding that much of the infrastructure necessary to self-aggregate data today can be purchased from third-party vendors, so that in practice, the experience of purchasing the lowest latency access to consolidated market data may be similar whether the market participant choose to use a competing consolidator or to self-aggregate new consolidated market data.

self-aggregation, which prevent other market participants from becoming self-aggregators themselves and thereby preserves self-aggregators as the only market participants with such high quality information. To the extent that it happens that self-aggregation is necessary in order to obtain the maximum possible latency advantages,[2415] the exclusive advantage this offers will be reduced, because, whereas today one must purchase proprietary data feeds in order to employ this methodology of self-aggregation, under the final rule, the end user can purchase consolidated market data and employ this methodology through the self-aggregator option.[2416] This has the effect of reducing the costs to employ such technology, because the fees for new consolidated market data will likely be lower than the fees for proprietary data. Thus, rather than institutionalizing the advantages enjoyed by current users of the self-aggregation methodology, we expect the self-aggregator option in the decentralized consolidation model will reduce the barriers to entry into this level of market data access for other market participants.

While the final rules will not eliminate levels of sophistication in the utilization of market data, it will likely reduce the cost of moving between levels.[2417] With lower costs to increase

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2415 See, e.g., NBIM Letter at 4 ("In our experience, therefore, broker/dealers that do not undertake data aggregation in-house, and do not use the fastest connectivity available, will in general not be consistently competitive. This does not preclude using third-party technology to do the data aggregation, as long as it is done in-house to avoid incremental latency.").

2416 This of course depends on the extent to which the end user finds the content of new core data a viable substitute for proprietary data.

2417 These costs are the costs discussed in moving along the continuum of market data utilization methods in Section V.B.2(c). Also, market participants may be able to improve their use of market data under the final rules even if they currently utilize proprietary market data, because they may be able to substitute new core data for proprietary data. If they do switch, the likely cost savings they will obtain may enable

sophistication, the information asymmetry between the two tiers of market data access, of those who can afford and choose to purchase proprietary data and those who do not, will be reduced. This may lead to new entrants into the market making, executing broker-dealer, and latency sensitive trading businesses.

One commenter stated that the rule would put "retail investors who subscribe to a competing consolidator at a disadvantage relative to those traders who can afford to self-aggregate and generate their own 'NBBO' more quickly than retail investors reliant on third parties to obtain the NBBO."[2418] The Commission disagrees that retail investors in particular would be put at a disadvantage compared to self-aggregators as a result of the rule. As discussed above,[2419] currently, retail investors typically access the market using display and per quote feeds, which are not competitive in terms of speed with typical market data feeds. Investors who use such feeds understand that it is not possible to compete on speed and make their investment decisions based on other kinds of strategies. Thus, differences measured in microseconds, even if they resulted from this rule, would be meaningless to retail investors at the moment when they are making investment or trading decisions. Furthermore, retail investors, like many professional investors, do not execute their own trades but instead leave that function to their broker-dealer. For example, retail broker-dealers route their customers' orders to exchanges, ATSs, or wholesalers, the latter of which may route the order to the exchanges itself. Once the order has reached such market participants, the execution decisions are made in a much more sophisticated fashion (and microsecond differences matter), but crucially, these players will be

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them to invest in other aspects of the data access process, thereby improving their ability to compete with more sophisticated market participants.

[2418] See Nasdaq Letter IV at 41.

[2419] See supra Section V.B.2(c).